

DATA 23700

Project

See individual deliverables for due dates.

Groups of four students work together to produce an original data analysis and share their findings in an oral defense and a polished project repository. Groups first identify a motivating problem or question and find a dataset to analyze in order to address that topic. Then, they apply analysis and visualization techniques learned throughout the quarter to uncover a story and answer questions about the relationships in that dataset. The analysis should be compelling, reproducible, and appropriate for the chosen dataset. Groups produce a project repository containing a literate-programming analysis notebook and supporting materials, and they defend their findings to the instructor during the last week of instruction.

Students will work in groups of four. Groups are self-selected and must be finalized by the dataset audit deadline. Students are expected to contribute equitably to their group's work. Each project repository must include a contribution statement documenting how the work was divided (see the specification for the Project Repository).

Deliverables

The project consists of four deliverables, each with its own specification document containing detailed requirements and grading criteria:

- **Dataset Audit:** an early check-in with the instructor or a TA to demonstrate that the group's chosen dataset can support their motivating question. See `1_dataset_audit.pdf`.
- **Written Proposal:** a short document motivating the problem and describing the dataset, submitted as a PDF to Gradescope. See `2_proposal.pdf`.
- **Project Defense:** an oral exam in which the group pitches their project to the instructor in private and answers questions about it, held during the last week of instruction, with slides submitted in addition. See `3_defense.pdf`.
- **Project Repository:** a zip archive containing the analysis notebook, a README, and any supplementary code, submitted to Gradescope by the project deadline. See `4_repository.pdf`.

Choice of dataset

Choice of dataset is one of the most important decisions a group will make about their project. The dataset should be one of interest to the group, and it should *not* be one that we provide to students during the quarter. Importantly, the chosen dataset needs to be sufficiently complex to support an original final project on a well considered question or problem. This means that groups should not find an already assembled dataset (e.g.,

collated by a data journalist) and merely reproduce a previous analysis. Similarly, open-ended exploratory analysis of a clean dataset (e.g., from Kaggle) is insufficient, because such projects tend to involve too little motivation, not enough original work with the data source, and insubstantial conceptual interpretation of the data.

Rather, groups should approach the project with a central problem or question in mind. In past offerings, high-quality projects have involved either: (1) finding a large, messy dataset with many variables and a complex data-generating process to analyze; or (2) sleuthing and fusion of multiple data sources to construct a dataset that can answer a targeted question. *It is important to choose a dataset that can support an analysis of sufficient depth to demonstrate skills acquired in the course and of sufficient interest to support a narrative about the analysis akin to a technical report or a piece of data journalism.*

The Dataset Audit deliverable is designed to help groups make this choice deliberately and get feedback before they commit. Groups are strongly encouraged to visit office hours during the first weeks of the quarter to workshop project ideas.

Grading

Each deliverable is evaluated according to the criteria in its specification document. See the course syllabus for how project scores contribute to the final grade.

Resubmissions. The Dataset Audit and Written Proposal may be resubmitted per the standard assignment resubmission policy (see syllabus). The Project Defense and Project Repository may *not* be resubmitted. The intent is that resubmissions on the audit and proposal give struggling groups a way to arrive at a viable project idea before it's too late in the quarter.

Late work. The project follows the course's standard late work policy: late submissions are not accepted, with the exceptions specified in the syllabus.

Academic honesty

Groups are expected to produce original work. Collaboration *within* a group is expected and encouraged; collaboration *between* groups is not permitted. All group members must contribute substantively to the project, and the contribution statement in the Project Repository must accurately reflect the division of labor. Misrepresenting one person's work as another's—or presenting AI-generated work as your own—is an academic honesty violation. The generative AI policy stated in the syllabus applies to the project as it does to other coursework.